SG No. 151 D_3^3 $P3_112$ [trigonal]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `3a`, site symmetry: `..2`

Table 1: Wyckoff bond: `3a@3a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ -Z]$	$\left[x,\ -x,\ \frac{1}{3}\right]$	`[1,-4]`
2	$[-X,\ 0,\ -Z]$	$\left[x,\ 2x,\ \frac{2}{3}\right]$	`[2,-6]`
3	$[0,\ -X,\ -Z]$	$[-2x,\ -x,\ 0]$	`[3,-5]`

Table 2: Wyckoff bond: `3b@3a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{1}{3}\right]$	`[1,4]`
2	$[X,\ 2X,\ 0]$	$\left[x,\ 2x,\ \frac{2}{3}\right]$	`[2,6]`
3	$[-2X,\ -X,\ 0]$	$[-2x,\ -x,\ 0]$	`[3,5]`

Table 3: Wyckoff bond: `6c@3a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ -x,\ \frac{1}{3}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[x,\ 2x,\ \frac{2}{3}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$[-2x,\ -x,\ 0]$	`[3]`
4	$[-Y,\ -X,\ -Z]$	$\left[x,\ -x,\ \frac{1}{3}\right]$	`[4]`
5	$[-X+Y,\ Y,\ -Z]$	$[-2x,\ -x,\ 0]$	`[5]`
6	$[X,\ X-Y,\ -Z]$	$\left[x,\ 2x,\ \frac{2}{3}\right]$	`[6]`

* Wyckoff site: `3b`, site symmetry: `..2`

Table 4: Wyckoff bond: `3a@3b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ -Z]$	$\left[x,\ -x,\ \frac{5}{6}\right]$	`[1,-4]`
2	$[-X,\ 0,\ -Z]$	$\left[x,\ 2x,\ \frac{1}{6}\right]$	`[2,-6]`
3	$[0,\ -X,\ -Z]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[3,-5]`

Table 5: Wyckoff bond: `3b@3b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ -X,\ 0]$	$\left[x,\ -x,\ \frac{5}{6}\right]$	`[1,4]`
2	$[X,\ 2X,\ 0]$	$\left[x,\ 2x,\ \frac{1}{6}\right]$	`[2,6]`
3	$[-2X,\ -X,\ 0]$	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[3,5]`

Table 6: Wyckoff bond: `6c@3b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[x,\, -x,\, \frac{5}{6}\right]$	`[1]`
2	$[-Y,\, X-Y,\, Z]$	$\left[x,\, 2x,\, \frac{1}{6}\right]$	`[2]`
3	$[-X+Y,\, -X,\, Z]$	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	`[3]`
4	$[-Y,\, -X,\, -Z]$	$\left[x,\, -x,\, \frac{5}{6}\right]$	`[4]`
5	$[-X+Y,\, Y,\, -Z]$	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	`[5]`
6	$[X,\, X-Y,\, -Z]$	$\left[x,\, 2x,\, \frac{1}{6}\right]$	`[6]`

* Wyckoff site: `6c`, site symmetry: `1`

Table 7: Wyckoff bond: `6a@6c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	`[1]`
2	$[-Y,\, X-Y,\, Z]$	$\left[-y,\, x-y,\, z+\frac{1}{3}\right]$	`[2]`
3	$[-X+Y,\, -X,\, Z]$	$\left[-x+y,\, -x,\, z+\frac{2}{3}\right]$	`[3]`
4	$[-Y,\, -X,\, -Z]$	$\left[-y,\, -x,\, \frac{2}{3}-z\right]$	`[4]`
5	$[-X+Y,\, Y,\, -Z]$	$\left[-x+y,\, y,\, \frac{1}{3}-z\right]$	`[5]`
6	$[X,\, X-Y,\, -Z]$	$[x,\, x-y,\, -z]$	`[6]`